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Beating Go and the road ahead for AI

We sit down with Deep Mind's David Silver speaking about Google's AlphaGo AI, its huge step towards achieving artificial intelligence and its future plans.



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Firstly, could you explain the significance of AlphaGo's victory over Lee Seedol? Why does a victory in Go matter so much?

David: Go represents a major milestone in AI research. Go is enormously complex, and such an intuitive game that so-called "brute force" search techniques are not sufficient. Until recently, computers have only played Go as well as amateurs. Go's simple rules and profound complexity resemble the real-world in many ways. Therefore we think Go is an important stepping stone towards a more general AI.

How does AlphaGo's victory in Go differ from Deep Blue's victory earlier. How do Chess and Go differ when it comes to being played by an AI?

David: The number of possible moves in chess is much lower. The search space in Go is vast – more than a googol times larger than chess (a number greater than there are atoms in the universe!) So traditional "brute force" AI methods, which construct a search tree over all possible sequences of moves, don't have a chance in Go. In addition, it's easier to evaluate who is ahead in chess

Human grandmasters handcrafted the chess knowledge used by Deep Blue

to evaluate chess positions. For AlphaGo, we did not program what strategy to use – instead it learnt for itself from hundreds of thousands of human expert games, and from millions more games of self-play. Finally, Deep Blue used a search algorithm that was based more on brute force – it evaluated thousands of times more positions than AlphaGo. Instead, AlphaGo searches much more smartly and selectively.

AlphaGo has two neural networks working inside it. Could you explain these and how they work?

David: One neural network, the "policy network," selects the next move to play. The other neural network, the "value network," predicts the winner of the game. Each neural network takes the board position and passes it through many computational layers with millions of tunable weights, to come up with its answer.

AlphaGo's gameplay has been described as aggressive in the tournament. How does it do that? Why is that significant? Is there no rule-based programming at play here?

David: AlphaGo always selects the move that maximises its chances of winning the game. This can sometimes result in aggressive moves, but can also sometimes result in seemingly quiet and defensive moves

How easy or difficult will it be to adapt AlphaGo to other activities?

David: The methods we've used for AlphaGo are general-purpose, so our hope is that one day they could be extended to help us address other challenges from making smartphone assistants more useful to helping scientists with some of society's toughest and most pressing problems, from climate modelling to complex disease analysis. That being said, it's very early days. We are decades away from human level AI.

If yes, does that make it difficult to scale or adapt AlphaGo to other applications?

David: This is the beginning, and there is a lot more work to be done before we can begin to apply these techniques to real world problems. But with the right developments it isn't impossible to have such implications.

Do you plan things like AlphaGo to be used for consumer applications, like in smartphone voice assistants?

David: Not necessarily AlphaGo, but the machine learning techniques will be a tool that helps us do what we already do, but better – whether that's instant translation, a smarter assistant on our phone that can plan a trip for us just by knowing what we like or by helping doctors to diagnose a disease much earlier.